Theorem—Fatou's lemma. Given a measure space $(\Omega, \mathcal{F}, \mu)$ and a set $X \in \mathcal{F}$, let $\{f_n\}$ be a sequence of $(\mathcal{F}, \mathcal{B}_{\bar{\mathbb{R}}_{\geq 0}})$-measurable non-negative functions $f_n : X \to [0, +\infty]$. Define the function $f : X \to [0, +\infty]$ by

$$f(x) = \liminf_{n\to\infty} f_n(x), \text{ for every } x \in X.$$

Then f is $(\mathcal{F}, \mathcal{B}_{\bar{\mathbb{R}}_{\geq 0}})$-measurable, and

$$\boxed{\int_X f\, d\mu \leq \liminf_{n\to\infty} \int_X f_n\, d\mu,}$$

where the integrals and the Limit inferior may be infinite.

$(\mathbb{R}^d, \mathcal{L}, \mu)$

$\mathbb{R}$

$|f_n| : \mathbb{R}^d \to [0, \infty)$

$|f| : \mathbb{R}^d \to [0, +\infty]$

$|f| = |c| = \lim_{n\to\infty} |f_n(x)| = \underline{\lim}_{n\to\infty} |f_n(x)|$

$\int_{\mathbb{R}^d} |c|\, d\mu \leq \underline{\lim}_{n\to\infty} \int_{\mathbb{R}^d} |f_n|\, d\mu \leq M_1$

$\Rightarrow \boxed{\infty \leq M_1}$ if $c \neq 0$

$\Rightarrow\Leftarrow$

M_∞

$c \in \mathbb{R}$